Evaluating Time Series Foundation Models for Remaining Useful Life Prediction: A Comparative Study of Chronos-1 and Chronos-2

Extended version kept as a research record. It complements the 8-page conference version with the full multi-seed grid (including the XGBoost head), re-run baselines, the statistical analysis and reproducibility notes (Sect. 4.5 and 6). Results: run of 2 November 2026, notebook `W09_FD001_MultiSeed_Stats.ipynb`.

Abstract. Knowing how long an industrial asset can keep operating before it fails, its Remaining Useful Life, is central to predictive maintenance. Deep models trained for this task, however, need large amounts of rare run-to-failure data and must be retrained whenever the monitored system changes. Foundation models pretrained on generic time series offer an alternative, since their representations can be reused as they are. This paper studies when such representations are useful for life prediction, by comparing the univariate Chronos-1 and the multivariate Chronos-2 on the NASA turbofan engine benchmark. Both models are kept frozen, and their outputs are passed to lightweight regressors under several pooling and sensor selection strategies. Over five random seeds, the best Chronos-2 configuration reaches an RMSE of 14.61 $\pm$ 0.17 cycles, improving on Chronos-1 (14.92) and coming close to a task-specific recurrent network trained under the same protocol (13.93 $\pm$ 0.90), without statistically significant difference. Keeping a diverse set of sensors also matters more than keeping only the most predictive ones (14.61 vs. 19.16).

Keywords: Time series foundation models, Predictive maintenance, Remaining useful life, Transfer learning, Chronos

1 Introduction

Predictive maintenance (PdM) aims to minimise unplanned downtime and maintenance costs [11]. At its core lies the estimation of the Remaining Useful Life (RUL), which allows engineers to move from reactive interventions to condition-driven strategies. Its impact extends beyond maintenance, since an unexpected failure quickly turns into production losses and contractual penalties.

Today, the most accurate RUL predictors are deep networks trained for one specific asset. Their strength is also their weakness: they require labelled run-to-failure trajectories, which are rare and costly to collect, and they must be retrained when the target system changes. Time Series Foundation Models (TSFMs) approach the problem from the opposite direction. Pretrained once on large and diverse corpora, they provide generic temporal representations that

can be reused without retraining the backbone [4]. Within Amazon's Chronos family, Chronos-1 is univariate and therefore ignores the dependencies between sensors, whereas Chronos-2 [3] processes several variables jointly through group attention. Whether, and under which design choices, these frozen representations are useful for RUL regression is still poorly understood.

This paper addresses this question through a controlled evaluation of frozen Chronos-1 and Chronos-2 representations on C-MAPSS FD001. Our contributions are:

1. a comparison of Chronos-1 and Chronos-2 under a protocol identical to a published Chronos-1 study [5], so that differences can be attributed to the backbone;
2. a systematic exploration of the design space of frozen embeddings (pooling strategy, regression head and feature selection);
3. evidence that feature diversity, rather than individual feature relevance, drives Chronos-2 performance.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the methodology. Section 4 presents experimental results. Section 5 discusses key findings, Section 6 gathers reproducibility notes, and Section 7 concludes the paper.

2 Related Work

This section reviews task-specific deep learning for RUL prediction, then introduces time series foundation models, and finally discusses how these models have been applied to RUL prediction.

Task-specific deep learning for RUL. Model-based approaches rely on physical degradation models, which are difficult to build for complex systems [11]; this difficulty has motivated data-driven alternatives, starting with recurrent networks [9]. Convolutional networks (CNNs) were then applied to sensor windows [13], and deeper 1-D CNNs reached an RMSE of 12.61 on FD001 [12], although convolutions only capture local temporal context. Long short-term memory networks (LSTMs) [10] overcome this limitation by modelling long-range dependencies and have become a standard for RUL prediction [18], sometimes combined with CNNs in hybrid architectures [17]. Whatever their architecture, however, these models are trained from scratch for one dataset and must be retrained when the system changes.

Time series foundation models. TSFMs were introduced precisely to avoid this repeated training. Pretrained on heterogeneous time series, they differ mainly in how they represent the input. Chronos [4] scales and quantises values into tokens and trains T5 language models with a cross-entropy loss, while TimesFM [6] is a decoder-only transformer operating on patches. MOIRAI [16] relies on any-variate attention to handle a variable number of channels. MOMENT [8], pretrained by masked reconstruction, explicitly targets representation learning. Finally, Chronos-2 [3] is an encoder that processes groups of related series jointly

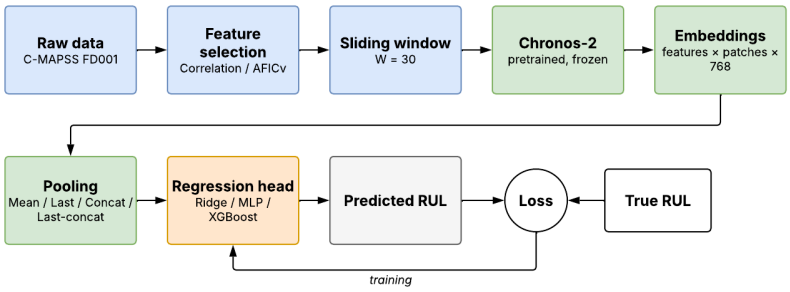


Fig. 1. Overview of the evaluated pipeline.

through group attention. Being pretrained for forecasting or reconstruction, all of them require a transfer step to be applied to RUL.

TSFMs for RUL prediction. Several recent works have explored such transfer. A first study [5] used frozen Chronos-1 embeddings, extracted sensor by sensor, with XGBoost and MLP heads on FD001; a task-specific LSTM nevertheless remained better (RMSE 14.32 vs. 14.92). Dintén and Zorrilla [7] followed a similar idea with MOMENT embeddings and classical regressors, showing benefits in few-shot settings. Concurrently with the present work, Abdouni et al. [1] went further by adding a trainable compress-and-flatten adapter on top of Chronos-2 and a Wide & Deep head fed with handcrafted features, reaching 9.97 on FD001 (10.68 without the handcrafted branch). To our knowledge, TimesFM and MOIRAI have not yet been evaluated for RUL on C-MAPSS. None of these studies examines how pooling, regression head and feature selection interact with a frozen multivariate backbone; this is the gap addressed here. For this reason, [5] serves as the primary baseline: it shares our preprocessing, window, RUL cap, heads and evaluation protocol, and is therefore the only reference for which the Chronos-1/Chronos-2 difference is not confounded by protocol differences.

3 Methodology

Building on this positioning, the evaluated pipeline comprises four steps (Fig. 1): problem formulation, feature selection, frozen embedding extraction and regression.

3.1 Problem Formulation

Given multivariate readings $\mathbf{X} = \{x_1, \ldots, x_T\}$, $x_t \in \mathbb{R}^d$, of an engine failing at cycle T, the target is the piecewise-linear RUL $y_t = \min(T - t,\ R_{\max})$ with $R_{\max} = 125$ [15]. A sliding window of $W = 30$ cycles is used to predict the RUL at its last cycle. This value is the standard choice for FD001 [12]; it is also the largest window compatible with the shortest test trajectory (31 cycles), so every test engine can be evaluated without padding.

Table 1. Effect of the two feature selection strategies on the number of input variables (FD001).

Step	Criterion	Input	Removed	Output
Correlation-based strategy				
Variance filter	variance < 0.001	24	7	17
Correlation filter	$\lvert\rho\rvert > 0.95$	17	1	**16**
AFICv strategy				
Importance ranking	90% cum. importance	24	16	**8**

3.2 Feature Selection

FD001 provides 24 raw variables (21 sensors and 3 operating settings), min-max normalised with statistics fitted on the training set. Two strategies are compared to select the model inputs (Table 1). The *correlation-based* strategy aims at a diverse, non-redundant set: it removes near-constant variables, then one variable of each highly correlated pair. In contrast, *AFICv* [2] focuses on individual relevance, keeping the smallest subset of variables that covers 90% of the cumulative gain importance of an XGBoost regressor fitted on the training set.

3.3 Embedding Extraction

The selected variables are then encoded by one of the two frozen backbones. *Chronos-1* (`chronos-t5-small`, 46M parameters) is univariate, so each sensor window $\mathbf{W}_{:,i} \in \mathbb{R}^W$ is encoded separately and averaged over time, and the d sensor embeddings are concatenated:

$$\mathbf{e} = \big[\,\text{MeanPool}(\text{Chronos-1}(\mathbf{W}_{:,i}))\,\big]_{i=1}^{d}. \tag{1}$$

By construction, this representation cannot capture cross-sensor dependencies. *Chronos-2* (`amazon/chronos-2`, 120M parameters) removes this limitation by encoding all d variables jointly through group attention. Its output is a tensor of shape (d, n_{patches}, d_{model}=768), which must be reduced to a vector $\mathbf{e} = \text{Pool}(\text{Chronos-2}(\mathbf{W}))$. Four pooling strategies are compared for this purpose: *mean* (average over variables and patches, 768 dimensions), *last* (last patch, averaged over variables, 768), *concat* (average over variables, patches concatenated, 3,072) and *last-concat* (last patch of each variable concatenated, 12,288).

3.4 Regression Heads and Evaluation

Finally, three heads map $\mathbf{e}$ to the estimate $\hat{y}_t$, as in [5]: a linear *Ridge* regression ($\alpha = 1$), a non-linear *MLP* (256 and 128 ReLU units, Adam, early stopping on 10% of the training windows) and *XGBoost* (100 trees, depth 5). The main text focuses on Ridge and MLP; XGBoost results, which proved sensitive to the

Table 2. Chronos-2 RMSE by pooling and head (correlation features). †Single run.

Pooling	Dim	Ridge	MLP
Mean	768	15.68	**14.61** ± 0.17
Last	768	**15.20**	15.21 ± 0.23
Concat	3,072	15.49	15.07 ± 0.23
Last-concat†	12,288	–	18.03

Table 3. Chronos-2 (mean pooling) by feature set and head.

Feat.	Head	RMSE	Score
Corr.	MLP	**14.61** ± 0.17	**358** ± 30
Corr.	Ridge	15.68	388
AFICv	MLP	19.16 ± 0.48	939 ± 129
AFICv	Ridge	19.03	772

library version (Sect. 6), are reported in Sect. 4.5. Only the head is trained. Performance is measured on the last window of each test engine by RMSE $= \sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}$ and by the NASA score $S = \sum_i s_i$, with $s_i = e^{-d_i/13} - 1$ if $d_i < 0$ and $e^{d_i/10} - 1$ otherwise, where $d_i = \hat{y}_i - y_i$ [14]; the score penalises late predictions more heavily, as they are the most dangerous in practice.

4 Experiments and Results

This section first describes the dataset and experimental setup, then analyses the impact of the pooling strategy, the regression head and the feature selection, and finally compares the best configuration with LSTM baselines and other Chronos-based models.

4.1 Experimental Setup

The experiments use C-MAPSS FD001 [15], which contains 100 training engines run to failure (20,631 cycles) and 100 test engines truncated before failure (13,096 cycles), with the true RUL provided for the latter, under one operating condition and one fault mode. Windowing yields 17,731 training and 10,196 test windows, and the last window of each of the 100 test engines is scored. FD001 is the smallest C-MAPSS subset (FD002 and FD004 contain 260 and 249 training engines), a small-data regime where transferable representations should help. Chronos embeddings are deterministic, as is the Ridge head. The MLP head and a task-specific baseline (two-layer LSTM with 64 units, trained end to end on the same windows with early stopping) are trained with five random seeds (42, 123, 456, 789, 1000), and results are reported as mean ± standard deviation. Embeddings are extracted on a T4 GPU.

4.2 Pooling Strategy and Regression Head

The first experiment identifies how the Chronos-2 output should be summarised (Table 2, correlation-based features).

Mean pooling with the MLP gives the lowest RMSE and is retained for the next experiments. Interestingly, the linear Ridge head prefers the last patch (15.20) to the mean (15.68), the opposite of the MLP.

Table 4. Comparison of LSTM baselines and Chronos-based models on FD001. Upper block: identical protocol (mean ± std over five seeds for re-trained models; *values reported in [5]). Lower block: trained adapter on Chronos-2, different protocol (values as reported).

Method	RMSE	Score
Task-specific LSTM (re-trained)	**13.93 ± 0.90**	378.6 ± 100.0
Task-specific LSTM*	14.32	**325.1**
Chronos-2 + MLP, mean (ours)	14.61 ± 0.17	357.9 ± 30.4
Chronos-1 + MLP*	14.92	641.6
Chronos-2 + Ridge, last (ours)	15.20	349.8
Chronos-1 + XGBoost*	15.93	379.0
Classical LSTM*	16.14	397.8
Chronos-2 + adapter, deep only [1]	10.68 ± 0.34	–
Chronos-2 + adapter, Wide&Deep [1]	9.97 ± 0.34	187.4

4.3 Feature Selection

With the pooling fixed, the second experiment compares the two feature sets across both heads (Table 3). The gap is large and consistent: whatever the head, correlation-based selection outperforms AFICv by at least 3.3 cycles, far beyond the seed-to-seed variability.

4.4 Comparison with LSTM and Chronos-Based Models

The best configuration can now be positioned against the LSTM baselines and the other Chronos-based models (Table 4). The upper block gathers models evaluated under the identical protocol, so that differences reflect the models themselves. The lower block reports the concurrent Chronos-2 models of [1], which use a different training pipeline and are given for reference.

4.5 Extended Results and Statistical Analysis

This subsection gathers all results of the multi-seed run, including the XGBoost head and baselines re-trained under the identical protocol, as a reference for future work.

To assess the significance of the differences, predictions were averaged over seeds and compared with the reference configuration (Chronos-2 + MLP, mean pooling, correlation features) on the 100 test engines, using 95% bootstrap confidence intervals (5,000 resamples) and paired Wilcoxon signed-rank tests on per-engine absolute errors (Table 7).

Three observations complement the main results. First, none of the task-specific baselines is significantly better than Chronos-2 + MLP: the confidence intervals of the differences all contain zero, which reflects the limited size of the FD001 test set (100 engines). Second, every AFICv configuration is significantly

Table 5. Full Chronos-2 grid on FD001 (RMSE, mean ± std over five seeds; Ridge and XGBoost are deterministic). Corr.: correlation-based features (16); AFICv: 8 features.

Feat.	Pooling	Dim	Ridge	MLP	XGBoost
Corr.	Mean	768	15.68	**14.61** ± 0.17	16.97
Corr.	Last	768	15.20	15.21 ± 0.23	17.11
Corr.	Concat	3,072	15.49	15.07 ± 0.23	15.87
AFICv	Mean	768	19.03	19.16 ± 0.48	20.36
AFICv	Last	768	18.49	19.40 ± 0.26	21.19
AFICv	Concat	3,072	19.39	18.59 ± 0.36	20.33

Table 6. Baselines re-trained under the identical protocol (five seeds). Chronos-1 uses mean pooling over time and correlation-based features.

Model	RMSE	Range	Score
XGBoost on raw windows (480 inputs)	**13.63**	–	339.3
Task-specific LSTM (2×64)	13.93 ± 0.90	13.10–15.37	378.6 ± 100.0
Chronos-2 + MLP, mean	14.61 ± 0.17	14.39–14.84	357.9 ± 30.4
Chronos-1 + Ridge	15.68	–	361.0
Chronos-1 + XGBoost	17.52	–	563.9
Chronos-1 + MLP	18.19 ± 0.32	17.93–18.75	685.5 ± 12.1

worse than the reference ($p \leq 0.033$ for all heads and poolings), which makes the feature diversity effect the most robust finding of the study. Third, the XGBoost head is consistently the weakest head on Chronos-2 embeddings, and Chronos-1 + MLP is clearly worse than Chronos-2 + MLP, although the difference only borders on significance (CI lower bound 0.10).

5 Discussion

This section interprets the results and their practical implications.

Pooling strategy. As seen in Table 2, compact mean pooling suits the MLP best, whereas the Ridge head performs best with the last patch, which reflects the most recent degradation state and is easier to exploit linearly. The MLP, in contrast, benefits from a summary of the whole window, and the very high-dimensional last-concat representation degrades it markedly. Pooling and head should therefore be chosen jointly.

Feature selection. The feature selection results point in the same direction. As Table 1 shows, AFICv keeps only half as many variables (8 vs. 16), retaining the individually most predictive ones, some of which are strongly correlated with each other, while discarding sensors that are weak in isolation yet provide context to Chronos-2's group attention. For a multivariate foundation model, diversity of the input group thus matters more than individual relevance.

Table 7. Statistical comparison with Chronos-2 + MLP (mean, correlation). RMSE of seed-averaged predictions, 95% bootstrap CI, difference in RMSE with its 95% CI, and paired Wilcoxon p-value.

Configuration	**RMSE**	**95% CI**	**ΔRMSE [95% CI]**	p
Chronos-2 + MLP, mean (ref.)	14.49	[12.47, 16.47]	–	–
XGBoost on raw windows	13.63	[11.37, 15.88]	−0.86 [−3.90, 2.27]	0.264
Task-specific LSTM	13.53	[11.21, 15.77]	−0.97 [−3.63, 1.63]	0.115
Chronos-2 + MLP, concat	14.80	[12.77, 16.70]	0.31 [−0.85, 1.45]	0.195
Chronos-2 + Ridge, last	15.21	[13.38, 17.00]	0.71 [−0.57, 2.03]	0.156
Chronos-2 + Ridge, mean	15.68	[13.74, 17.53]	1.19 [0.19, 2.23]	0.040
Chronos-1 + Ridge	15.68	[13.82, 17.44]	1.19 [−1.54, 3.86]	0.128
Chronos-1 + MLP	17.94	[14.98, 20.65]	3.45 [0.10, 6.91]	0.552
Chronos-2 + MLP, mean, AFICv	19.01	[15.98, 21.75]	4.52 [2.44, 6.42]	0.007
Chronos-2 + Ridge, mean, AFICv	19.03	[16.37, 21.61]	4.54 [2.59, 6.41]	<0.001

Chronos-2 vs. Chronos-1. Under the same protocol, Chronos-2 + MLP improves on both Chronos-1 + MLP (14.61 vs. 14.92) and Chronos-1 + XGBoost (15.93), with a markedly better NASA score (357.9 vs. 641.6). This consistent trend supports joint multivariate encoding over per-sensor concatenation.

Why use a frozen foundation model? Compared with the task-specific LSTM, the remaining gap is small (14.61 vs. 13.93) and not statistically significant (paired Wilcoxon test on the 100 test engines, $p = 0.12$); the frozen approach is also much more stable across seeds (± 0.17 vs. ± 0.90). The frozen approach, however, only trains a lightweight head on embeddings that are computed once, instead of designing and training a recurrent architecture end to end, and the same backbone and embeddings can be reused across assets.

The adapter-based Chronos-2 models of [1] clearly outperform ours (9.97 vs. 14.61), but the two studies answer different questions, and our results remain informative for three reasons. First, their gain comes from components trained specifically for RUL: a learned adapter, a Wide & Deep head and handcrafted features. Ours involves none of these and measures what the frozen representation alone provides, so the gap between the two (14.61 vs. 10.68 without handcrafted features) quantifies the value of task-specific adaptation. Second, our findings concern choices made upstream of any head and thus transfer to such pipelines: the compress-and-flatten adapter is itself a learned pooling, consistent with our finding that pooling, like sensor diversity, strongly affects performance. Third, the frozen approach is the cheapest to deploy across many asset types with little labelled data. The two are thus complementary.

Limitations. These conclusions are limited to FD001, which involves a single operating condition.

6 Reproducibility Notes

The following points are recorded to ease the reproduction and extension of this study.

Software versions. The multi-seed run used xgboost 3.4.1, scikit-learn 1.6.1, PyTorch 2.11.0 and chronos-forecasting 2.3.2 (`amazon/chronos-2`, 119.5M parameters).

Sensitivity of XGBoost to the library version. With identical embeddings (the Ridge head gives exactly the same RMSE, 15.68), XGBoost on mean-pooled embeddings reached 15.62 in an earlier run (March 2026) and 16.97 in this run. The difference is attributed to changes in the default behaviour of newer xgboost releases. For the same reason, the AFICv subset changed slightly (sensor 7 replaces sensor 20), while its size (8 variables) is unchanged. XGBoost results should therefore always be reported together with the library version.

Chronos-1 discrepancy. The re-run Chronos-1 + MLP (18.19) is much worse than the value reported in [5] (14.92), whereas Chronos-1 + Ridge (15.68) is in line with the other configurations. Differences in the MLP training (early stopping) and in the embedding aggregation are the most likely causes; this point should be clarified before reusing the re-run Chronos-1 values.

LSTM training. A first LSTM implementation trained on unscaled targets diverged for one seed (RMSE 40.4). Scaling the targets to $[0, 1]$ by $R_{\max}$, clipping gradients (norm 1.0) and halving the learning rate on validation plateaus solved the problem (13.93 ± 0.90).

Pooling memory. The last-concat embeddings (12,288 dimensions) exceeded the 12.7 GB RAM of a standard Colab session when cached for the training windows, which is why this pooling was only evaluated once.

7 Conclusion

This paper evaluated frozen Chronos-1 and Chronos-2 representations for RUL prediction on C-MAPSS FD001 under a controlled protocol. The best Chronos-2 configuration (mean pooling, MLP head, correlation-based features) reaches an RMSE of 14.61 ± 0.17 over five seeds, improves on Chronos-1 and comes close to a task-specific LSTM trained under the same protocol (13.93 ± 0.90), while training only a lightweight head. Beyond these figures, two lessons emerge: feature diversity is essential for Chronos-2's group attention, and pooling and head interact strongly enough to be selected jointly. Building on these results, future work will address lightweight adaptation and fine-tuning with a RUL-specific loss, the multi-condition subsets FD002–FD004, and a comparison with other TSFMs such as TimesFM, MOIRAI and MOMENT under the same protocol.

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